

MySQL Assignment 2 -Querying Data

Clauses & Operators:

1. DISTINCT VALUES:

A query to retrieve distinct salaries from the Employees table.

```
SELECT DISTINCT salary FROM Employees;
```

salary
52000.00
48000.00
60000.00
55000.00
47000.00
51000.00
65000.00
49000.00

2. ALIAS (AS):

* Provide aliases for the "age" and "salary" columns as "Employee_Age" and "Employee_Salary", respectively.

```
SELECT age AS Employee_Age, salary AS Employee_Salary  
FROM Employees;
```

Employee_Age	Employee_Salary
29	52000.00
34	48000.00
41	60000.00
26	55000.00
38	47000.00
30	51000.00
45	65000.00

3. WHERE CLAUSE & OPERATORS:

* Retrieve employees with a salary greater than ₹50000 and hired before 2016-01-01.

```
SELECT * FROM Employees  
WHERE salary > 50000 AND hire_date < '2016-01-01';
```

employee_id	employee_name	gender	age	hire_date	designation	department_id	location_id	salary
1	Arun Kumar	M	29	2012-05-14	Data Analyst	1	1	52000.00
3	Rahul Singh	M	41	2010-03-10	Data Scientist	3	3	60000.00
7	Suresh Iyer	M	45	2011-06-30	Business Analyst	5	1	65000.00
11	Arjun Mehta	M	32	2011-04-12	Data Analyst	11	1	52000.00
13	Ravi Prasad	M	40	2015-09-23	Data Scientist	3	6	60000.00
17	Sanjay Gupta	M	44	2012-08-21	Business Analyst	19	8	65000.00
21	Rohan Das	M	37	2013-10-15	Data Scientist	14	12	58000.00

* Find the employee whose designation is missing and fill it with "Data Scientist".

```
SELECT employee_name, designation FROM Employees  
WHERE designation IS NULL;
```

	employee_name	designation
►	Rahul Singh	NULL
	Anjali Nair	NULL
	Divya Patel	NULL
	Ravi Prasad	NULL
	Meera Iyer	NULL

```
UPDATE Employees set designation='Data Scientist'  
WHERE designation IS NULL OR designation = '';
```

```
SELECT employee_name, designation FROM Employees  
WHERE designation = 'Data Scientist';
```

employee_name	designation
Rahul Singh	Data Scientist
Anjali Nair	Data Scientist
Divya Patel	Data Scientist
Ravi Prasad	Data Scientist
Meera Iyer	Data Scientist

1. ORDER BY:

- ```
SELECT * FROM Employees
ORDER BY department_id, salary DESC;
```

| employee_id | employee_name | gender | age | hire_date  | designation       | department_id | location_id | salary   |
|-------------|---------------|--------|-----|------------|-------------------|---------------|-------------|----------|
| 9           | Rohan Gupta   | M      | 33  | 2016-08-19 | Financial Analyst | 1             | 4           | 70000.00 |
| 4           | Meena Reddy   | F      | 26  | 2018-11-05 | Financial Analyst | 1             | 4           | 55000.00 |
| 1           | Arun Kumar    | M      | 29  | 2012-05-14 | Data Analyst      | 1             | 1           | 52000.00 |
| 56          | Pallavi Nair  | F      | 31  | 2018-06-05 | Data Scientist    | 2             | 17          | 51000.00 |
| 16          | Meera Iyer    | F      | 31  | 2018-06-05 | Data Scientist    | 2             | 7           | 51000.00 |
| 2           | Priya Sharma  | F      | 34  | 2015-07-22 | Business Analyst  | 2             | 2           | 48000.00 |
| 10          | Divya Patel   | F      | 27  | 2014-02-07 | Data Scientist    | 2             | 5           | 45000.00 |
| 3           | Rahul Singh   | M      | 41  | 2010-03-10 | Data Scientist    | 3             | 3           | 60000.00 |
| 13          | Ravi Prasad   | M      | 40  | 2015-09-23 | Data Scientist    | 3             | 6           | 60000.00 |
| 8           | Kavitha Menon | F      | 28  | 2019-04-25 | Data Analyst      | 3             | 3           | 49000.00 |
| 53          | Sandeep Reddy | M      | 40  | 2016-09-23 | Data Scientist    | 3             | 14          | 48000.00 |

- Display the first 5 employees hired in the year 2018.

```
SELECT * FROM Employees
WHERE YEAR(hire_date) = '2018'
ORDER BY hire_date LIMIT 5;
```

[illegible]

### 3. AGGREGATE FUNCTIONS:

- Calculate the sum of all salaries in the Finance department.

```
SELECT department_id,SUM(salary) FROM Employees
WHERE department_id=1
GROUP BY department_id;
```

| department_id | SUM(salary) |
|---------------|-------------|
| 1             | 177000.00   |

- Find the minimum age among all employees.

```
SELECT MIN(age) FROM Employees;
```

| MIN(age) |
|----------|
| 25       |

### 4. GROUP BY:

- List the maximum salary for each location.

| location_id | MAX(salary) |
|-------------|-------------|
| 1           | 70000.00    |
| 2           | 58000.00    |
| 3           | 70000.00    |
| 4           | 70000.00    |
| 5           | 58000.00    |
| 6           | 62000.00    |
| 7           | 65000.00    |
| 8           | 65000.00    |
| 9           | 55000.00    |
| 10          | 70000.00    |

```
SELECT location_id,MAX(salary) FROM Employees
GROUP BY location_id;
```

- Calculate the average salary for each designation containing the word 'Analyst'.

```
SELECT Designation, AVG(salary) FROM Employees
WHERE designation like '%Analyst%'
GROUP BY designation;
```

| Designation       | AVG(salary)  |
|-------------------|--------------|
| Data Analyst      | 40666.666667 |
| Business Analyst  | 51333.333333 |
| Financial Analyst | 63687.500000 |
| Research Analyst  | 53692.307692 |

## 5. HAVING:

- Find departments with less than 3 employees.

```
SELECT department_id, count(department_id) FROM Employees
GROUP BY department_id
HAVING count(department_id) > 3;
```

| department_id | count(department_id) |
|---------------|----------------------|
| 2             | 4                    |
| 3             | 5                    |
| 5             | 5                    |
| 6             | 6                    |
| 7             | 4                    |
| 11            | 10                   |
| 12            | 11                   |
| 13            | 14                   |
| 14            | 7                    |
| 15            | 12                   |

- Find locations with female employees whose average age is below 30.

```
SELECT Location_id, AVG(age) AS Avg_Age FROM Employees
WHERE gender = 'F'
GROUP BY Location_id
HAVING Avg_Age < 30;
```

| Location_id | Avg_Age |
|-------------|---------|
| 2           | 29.8571 |
| 3           | 29.5000 |
| 4           | 28.2500 |
| 5           | 27.0000 |
| 6           | 29.0000 |
| 8           | 28.0000 |
| 9           | 27.3333 |
| 11          | 28.5000 |
| 15          | 26.7500 |
| 17          | 29.8000 |



## Joins:

### 1. INNER JOIN:

- List employee names, their designations, and department names where employees are assigned to a department.

```
SELECT e.Employee_name, e.Designation, d.Department_name From Employees e
INNER JOIN departments d ON e.department_id = d.department_id;
```

| Employee_name | Designation       | Department_name   |
|---------------|-------------------|-------------------|
| Vivek Singh   | Data Scientist    | Data Scientist    |
| Ritika Sharma | Data Scientist    | Data Scientist    |
| Arun Kumar    | Data Analyst      | Finance           |
| Meena Reddy   | Financial Analyst | Finance           |
| Rohan Gupta   | Financial Analyst | Finance           |
| Divya Nair    | Financial Analyst | Financial Analyst |
| Vivek Sharma  | Financial Analyst | Financial Analyst |

### 2. LEFT JOIN:

- List all departments along with the total number of employees in each department, including departments with no employees.

```
SELECT d.Department_name, COUNT(e.employee_id) AS Total_Employees FROM Departments d
LEFT JOIN Employees e ON d.department_id = e.department_id
GROUP BY d.Department_name;
```

| Department_name   | Total_Employees |
|-------------------|-----------------|
| Administration    | 0               |
| Business Analyst  | 11              |
| Customer Support  | 0               |
| Data Analyst      | 10              |
| Data Scientist    | 7               |
| Distribution      | 0               |
| Finance           | 3               |
| Financial Analyst | 14              |
| Human Resources   | 4               |
| IT                | 5               |
| Legal             | 0               |
| Manager           | 6               |
| Marketing         | 3               |
| Operations        | 5               |
| Production        | 0               |
| Research          | 6               |

### 3. RIGHT JOIN:

- Display all locations along with the names of employees assigned to each location. If no employees are assigned to a location, display NULL for employee name.

```
SELECT l.Location_name, e.Employee_name FROM Employees e
RIGHT JOIN location l ON e.location_id = l.location_id;
```

| Location_name | Employee_name  |
|---------------|----------------|
| Delhi         | Ramesh Yadav   |
| Delhi         | Arjun Reddy    |
| Goa           | NULL           |
| Hyderabad     | Rahul Singh    |
| Hyderabad     | Kavitha Menon  |
| Hyderabad     | Rajesh Singh   |
| Hyderabad     | Rakesh Yadav   |
| Hyderabad     | Vivek Rao      |
| Hyderabad     | Meera Gupta    |
| Indore        | Ajay Kumar     |
| Indore        | Anand Krishnan |
| Indore        | Sandeep Reddy  |
| Indore        | Kiran Bhat     |

```
SELECT l.Location_name, GROUP_CONCAT(e.Employee_name) AS Employee_Name FROM Employees e
RIGHT JOIN location l ON e.location_id = l.location_id
GROUP BY l.Location_name;
```

| Location_name | Employee_Name                                      |
|---------------|----------------------------------------------------|
| Ahmedabad     | Vivek Sharma,Ashok Jain,Priyanka Joshi,Lakshm...   |
| Bangalore     | Priya Sharma,Anjali Nair,Sneha Kapoor,Nisha S...   |
| Bhopal        | Rohan Das,Pradeep Yadav,Anil Kumar,Aarti Jos...    |
| Chandigarh    | Neha Verma,Ritika Sharma,Ritu Sharma,Naveen...     |
| Chennai       | Arun Kumar,Suresh Iyer,Arjun Mehta,Harish Pa...    |
| Coimbatore    | Sanjay Gupta,Pooja Patel,Kiran Bhat,Deepa Kri...   |
| Delhi         | Vikram Das,Divya Patel,Karthik Reddy,Vignesh I...  |
| Goa           | NULL                                               |
| Hyderabad     | Rahul Singh,Kavitha Menon,Rajesh Singh,Rakes...    |
| Indore        | Ajay Kumar,Anand Krishnan,Sandeep Reddy,Kir...     |
| Jaipur        | Anitha Menon,Sonia George,Suraj Pillai,Manoj Pi... |
| Kochi         | Ravi Prasad,Aarti Joshi,Shreya Kapoor,Neha Ve...   |
| Kolkata       | Meera Iyer,Naveen Kumar,Anupam Das,Ajay K...       |
| Mumbai        | Meena Reddy,Rohan Gupta,Divya Nair,Pallavi G...    |

### 4. CROSS JOIN

- Show all possible combinations of departments and locations.

```
SELECT d.Department_name, l.Location_name FROM departments d
CROSS JOIN location l;
```

| Department_name  | Location_name |
|------------------|---------------|
| Business Analyst | Bhopal        |
| Business Analyst | Bangalore     |
| Business Analyst | Ahmedabad     |
| Customer Support | Trivandrum    |
| Customer Support | Patna         |
| Customer Support | Noida         |
| Customer Support | Mumbai        |
| Customer Support | Kolkata       |
| Customer Support | Kochi         |
| Customer Support | Jaipur        |
| Customer Support | Indore        |
| Customer Support | Hyderabad     |
| Customer Support | Goa           |
| Customer Support | Delhi         |

## 5. SELF JOIN:

- Show pairs of employees working in the same department, excluding self-pairs.

```
SELECT e1.Employee_name, e2.Employee_name, e1.Department_id
FROM Employees e1
JOIN Employees e2 ON e1.Department_id = e2.Department_id
AND e1.Employee_id <> e2.Employee_id -- exclude self pairs
ORDER BY e1.Department_id, e1.Employee_id;
```

| Employee_name | Employee_name | Department_id |
|---------------|---------------|---------------|
| Arun Kumar    | Meena Reddy   | 1             |
| Arun Kumar    | Rohan Gupta   | 1             |
| Meena Reddy   | Arun Kumar    | 1             |
| Meena Reddy   | Rohan Gupta   | 1             |
| Rohan Gupta   | Arun Kumar    | 1             |
| Rohan Gupta   | Meena Reddy   | 1             |
| Priya Sharma  | Divya Patel   | 2             |
| Priya Sharma  | Meera Iyer    | 2             |
| Priya Sharma  | Pallavi Nair  | 2             |
| Divya Patel   | Priya Sharma  | 2             |
| Divya Patel   | Meera Iyer    | 2             |
| Divya Patel   | Pallavi Nair  | 2             |
| Meera Iyer    | Priya Sharma  | 2             |
| Meera Iyer    | Divya Patel   | 2             |
| Meera Iyer    | Pallavi Nair  | 2             |
| Pallavi Nair  | Priya Sharma  | 2             |



## Windows function:

- Write a window function query to rank employees by salary using rank()

```
SELECT Employee_name, Designation,Salary, RANK() OVER(ORDER BY Salary DESC) AS Emp_Rank
From Employees ORDER BY Emp_Rank;
```

| Employee_name | Designation       | Salary   | Emp_Rank |
|---------------|-------------------|----------|----------|
| Rohan Gupta   | Financial Analyst | 70000.00 | 1        |
| Vivek Sharma  | Financial Analyst | 70000.00 | 1        |
| Harish Patel  | Financial Analyst | 70000.00 | 1        |
| Vivek Rao     | Financial Analyst | 70000.00 | 1        |
| Karthik Iyer  | Financial Analyst | 70000.00 | 1        |
| Rajesh Singh  | Financial Analyst | 68000.00 | 6        |
| Rajesh Singh  | Financial Analyst | 68000.00 | 6        |
| Aarti Sharma  | Financial Analyst | 68000.00 | 6        |
| Suresh Iyer   | Business Analyst  | 65000.00 | 9        |
| Sanjay Gupta  | Business Analyst  | 65000.00 | 9        |
| Pradeep Yadav | Research Analyst  | 65000.00 | 9        |
| Anupam Das    | Business Analyst  | 65000.00 | 9        |
| Rohit Gupta   | Business Analyst  | 65000.00 | 9        |
| Pradeep Yadav | Research Analyst  | 65000.00 | 9        |
| Pradeep Yadav | Research Analyst  | 65000.00 | 9        |
| Mohit Verma   | Manager           | 64000.00 | 16       |

- Write a window function query to rank employees by salary within each department using DENSE\_RANK()

```
SELECT Employee_Name, Designation,Department_Name, Salary,
DENSE_RANK() OVER(PARTITION BY E.Department_id ORDER BY Salary DESC) AS Emp_Dense_Rank
From Employees E INNER JOIN Departments D on E.Department_id= D.Department_id;
```

| Employee_Name | Designation       | Department_Name | Salary   | Emp_Dense_Rank |
|---------------|-------------------|-----------------|----------|----------------|
| Rohan Gupta   | Financial Analyst | Finance         | 70000.00 | 1              |
| Meena Reddy   | Financial Analyst | Finance         | 55000.00 | 2              |
| Arun Kumar    | Data Analyst      | Finance         | 52000.00 | 3              |
| Meera Iyer    | Data Scientist    | Human Resources | 51000.00 | 1              |
| Pallavi Nair  | Data Scientist    | Human Resources | 51000.00 | 1              |
| Priya Sharma  | Business Analyst  | Human Resources | 48000.00 | 2              |
| Divya Patel   | Data Scientist    | Human Resources | 45000.00 | 3              |
| Rahul Singh   | Data Scientist    | IT              | 60000.00 | 1              |
| Ravi Prasad   | Data Scientist    | IT              | 60000.00 | 1              |
| Kavitha Menon | Data Analyst      | IT              | 49000.00 | 2              |
| Sandeep Reddy | Data Scientist    | IT              | 48000.00 | 3              |
| Sneha Pillai  | Data Scientist    | IT              | 20000.00 | 4              |
| abcdefg       | Data Entry        | IT              | 12500.00 | 5              |
| Anjali Nair   | Data Scientist    | Marketing       | 51000.00 | 1              |
| Pooja Patel   | Data Scientist    | Marketing       | 45000.00 | 2              |
| Priya Patel   | Data Scientist    | Marketing       | 15000.00 | 3              |

- Write a window function query, Running total salary by department

```
SELECT Employee_Name, Designation, Department_Name, Salary, SUM(Salary) OVER(PARTITION BY E.Department_id) AS Tot_Salary
FROM Employees E INNER JOIN Departments D ON E.Department_id= D.Department_id;
```

| Employee_Name | Designation       | Department_Name | Salary   | Tot_Salary |
|---------------|-------------------|-----------------|----------|------------|
| Arun Kumar    | Data Analyst      | Finance         | 52000.00 | 177000.00  |
| Meena Reddy   | Financial Analyst | Finance         | 55000.00 | 177000.00  |
| Rohan Gupta   | Financial Analyst | Finance         | 70000.00 | 177000.00  |
| Priya Sharma  | Business Analyst  | Human Resources | 48000.00 | 195000.00  |
| Divya Patel   | Data Scientist    | Human Resources | 45000.00 | 195000.00  |
| Meera Iyer    | Data Scientist    | Human Resources | 51000.00 | 195000.00  |
| Pallavi Nair  | Data Scientist    | Human Resources | 51000.00 | 195000.00  |
| Rahul Singh   | Data Scientist    | IT              | 60000.00 | 249500.00  |
| Kavitha Menon | Data Analyst      | IT              | 49000.00 | 249500.00  |
| Ravi Prasad   | Data Scientist    | IT              | 60000.00 | 249500.00  |
| Sandeep Reddy | Data Scientist    | IT              | 48000.00 | 249500.00  |
| Sneha Pillai  | Data Scientist    | IT              | 20000.00 | 249500.00  |
| abcdefg       | Data Entry        | IT              | 12500.00 | 249500.00  |

```
SELECT Employee_Name, Designation, Department_Name, Salary,
SUM(Salary) OVER (PARTITION BY E.Department_id ORDER BY Salary ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW) AS Tot_Salary
FROM Employees E INNER JOIN Departments D ON E.Department_id = D.Department_id;
```

| Employee_Name | Designation       | Department_Name | Salary   | Tot_Salary |
|---------------|-------------------|-----------------|----------|------------|
| Arun Kumar    | Data Analyst      | Finance         | 52000.00 | 52000.00   |
| Meena Reddy   | Financial Analyst | Finance         | 55000.00 | 107000.00  |
| Rohan Gupta   | Financial Analyst | Finance         | 70000.00 | 177000.00  |
| Divya Patel   | Data Scientist    | Human Resources | 45000.00 | 45000.00   |
| Priya Sharma  | Business Analyst  | Human Resources | 48000.00 | 93000.00   |
| Meera Iyer    | Data Scientist    | Human Resources | 51000.00 | 144000.00  |
| Pallavi Nair  | Data Scientist    | Human Resources | 51000.00 | 195000.00  |
| abcdefg       | Data Entry        | IT              | 12500.00 | 12500.00   |
| Sneha Pillai  | Data Scientist    | IT              | 20000.00 | 32500.00   |
| Sandeep Reddy | Data Scientist    | IT              | 48000.00 | 80500.00   |
| Kavitha Menon | Data Analyst      | IT              | 49000.00 | 129500.00  |
| Rahul Singh   | Data Scientist    | IT              | 60000.00 | 189500.00  |
| Ravi Prasad   | Data Scientist    | IT              | 60000.00 | 249500.00  |